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Introduction to Recommender Systems

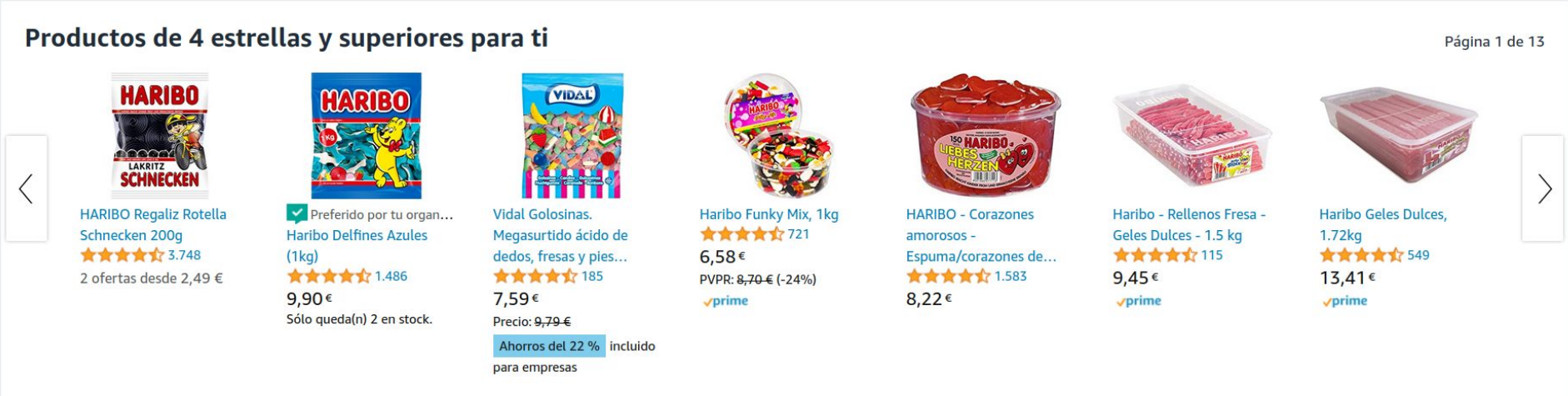
Javier Parapar López

What is a Recommender System?

- A working definition:

  «The goal of a recommender system is to generate meaningful recommendations to a collection of users for items or products that might interest them.»

amazon Spotify NETFLIX zalando



Example: e-commerce recommendations

ads Home My Books Browse ▾ Community ▾ Search books

Recommendations

Here are some books we think you'll love. How to improve your recommendations...

View: covers | list

YOUR INTEREST IN FANTASY

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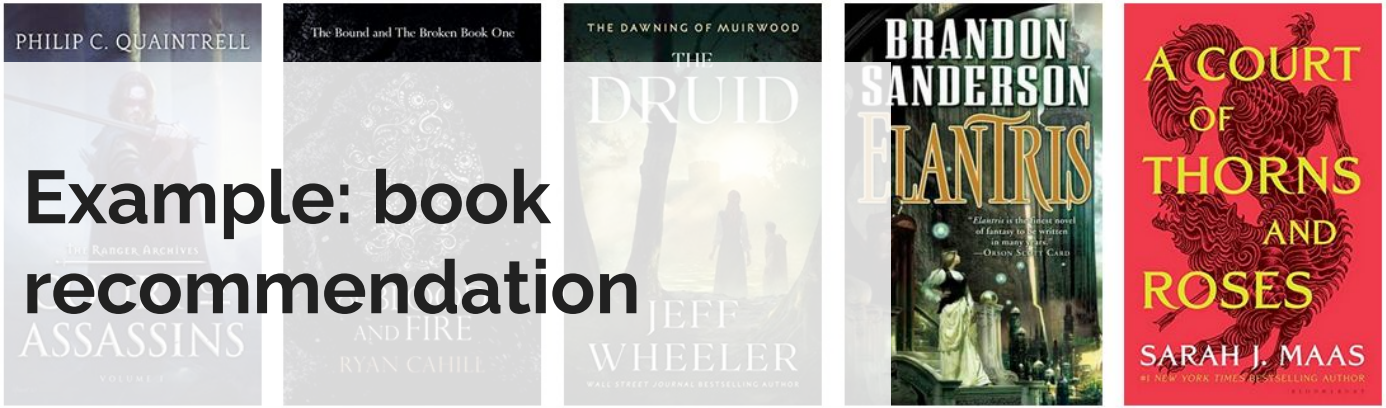


Example: book recommendation

Want to Read ★★★★★ Not interested

Want to Read ★★★★★ Not interested

Want to Read ★★★★★ Not interested

Want to Read ★★★★★ Not interested

Want to Read ★★★★★ Not interested

Recommendations by Shelf

overview

read

Tip: Create a new bookshelf of rel
that you've enjoyed or want to rea
recommendations based on those

Add a Shelf:

Recommendations by Genre

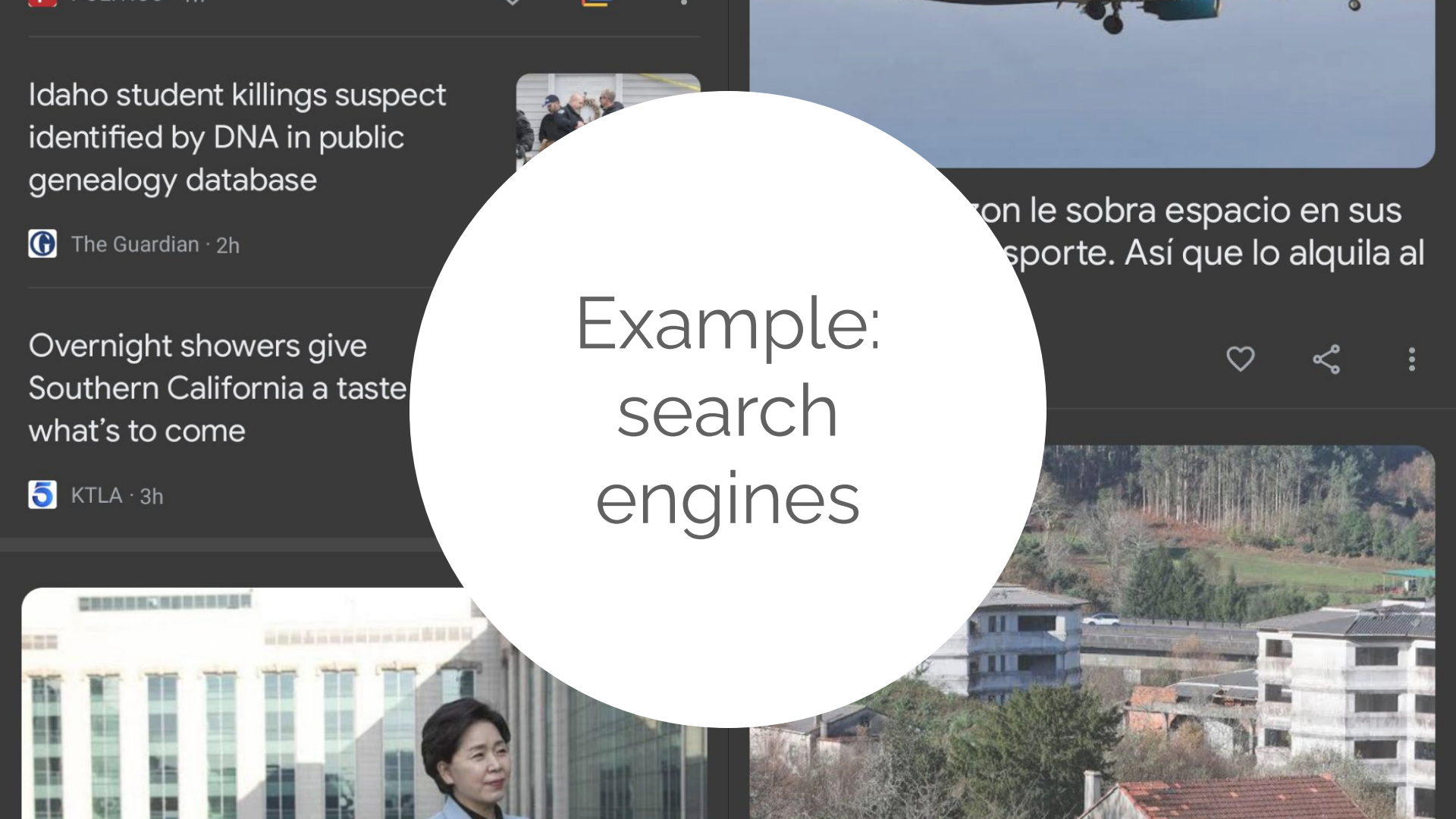
Fantasy

Fiction

Mystery

Science Fiction

Example: search engines



Example: top lists

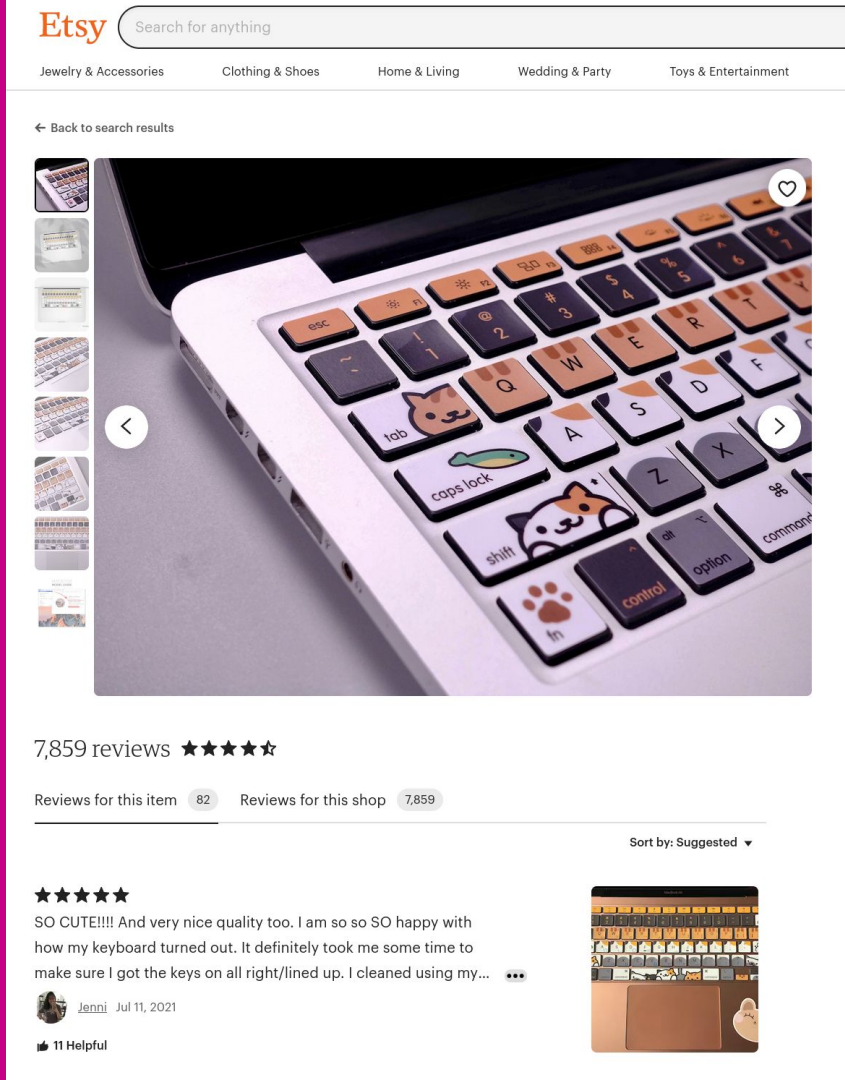
- Bestseller lists/charts for movies, music, websites, are a way of guiding users to items that they are *likely* to like
  - because many people seem to like them

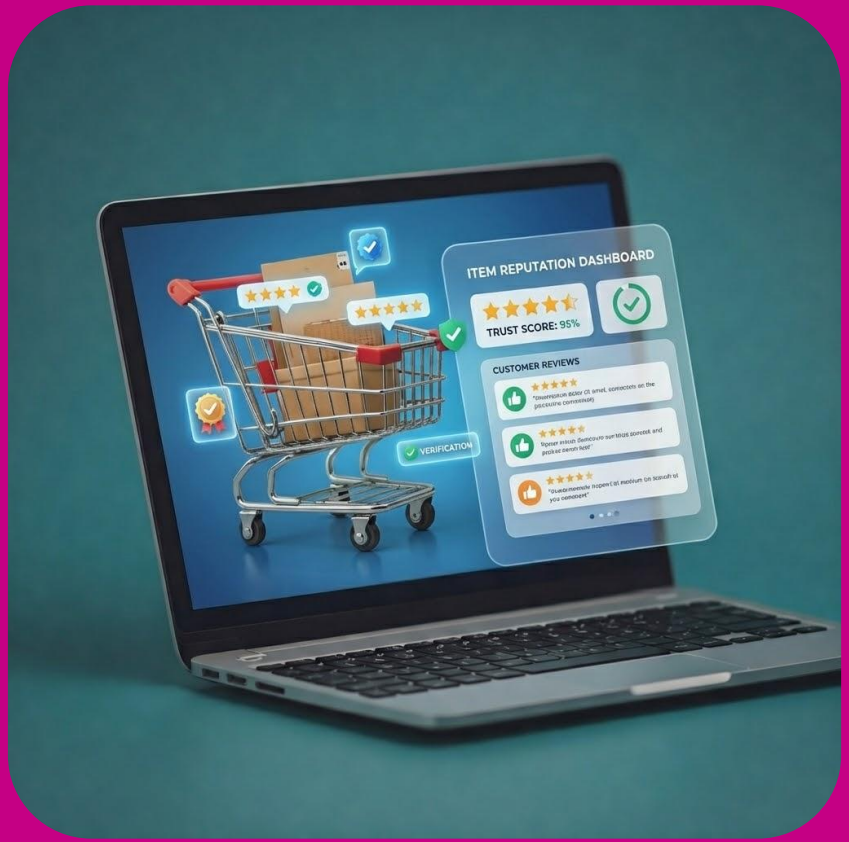
Quiz:

Any other recommender systems you've come across?

MACH

Reputation vs. Recommender systems





Reputation Systems

- Mechanisms to evaluate trustworthiness, reliability, or quality within a community or platform.
- Aggregate feedback and behavior into scores or ratings.
  - E-commerce: Amazon, eBay, Airbnb.
  - Social Media: Reddit, Stack Overflow.
  - Freelance Platforms: Upwork, Fiverr.
  - Gaming: Player behavior systems (e.g., League of Legends).

Reputation vs. Recommendation

- ✓ Both based on users' past reports
- ✓ Fundamental goal of both is to reduce a user's uncertainty about her satisfaction with a particular activity.

Reputation vs. Recommendation

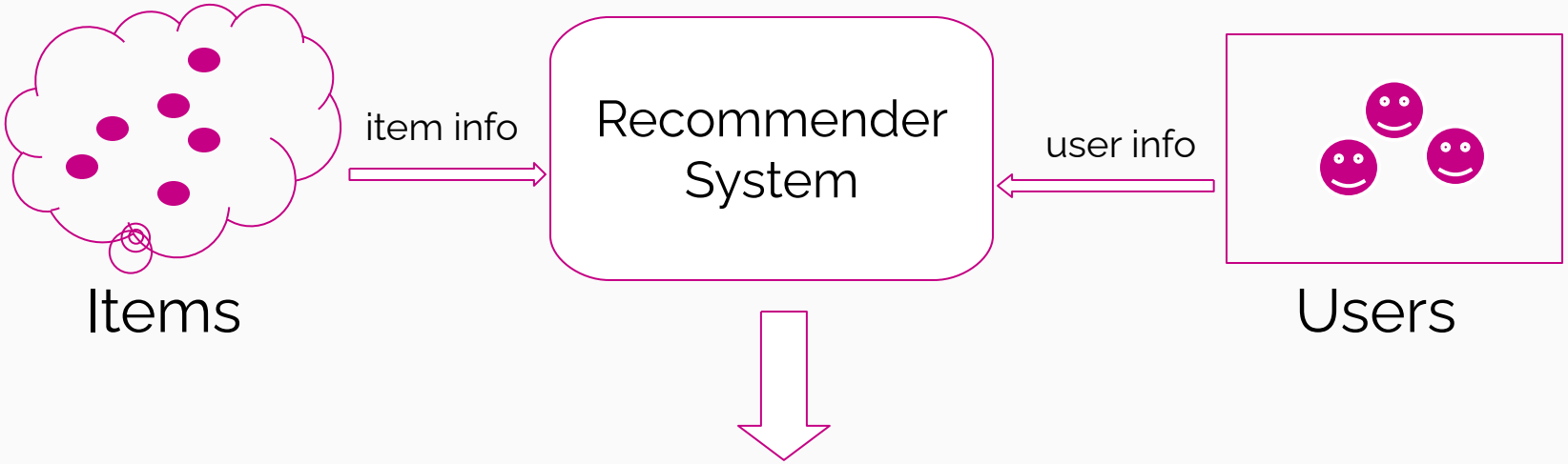
- Active agents vs. passive “items”
- Predicting future satisfaction vs. inducing appropriate actions
- Summarizing information (about an agent) vs. selecting from a group (of items.)

Outline of course

- Eliciting feedback/recommendation inputs
- Collaborative filtering algorithms (user-user, item-item, matrix-based models)
- Implementation detail and architectures
- Interface alternatives and effects
- Evaluation of Recommender Systems
- Anonymity and privacy issues

Coursework and evaluation

- Practical Assignments (5pts)
  - 40% min
- Final Exam (5pts)
  - 40% min
  - You can get additional +1 points by studying and presenting (10m) an ACM RecSys 2025 paper



Recommend item i to user u

The Design Space

- Major elements of the technical design space:
  - **Domain**: defines the set of items to recommend (e.g., movies, products, services).
  - **Identity Management:** mechanisms to identify users and their preferences (e.g., accounts, sessions).
  - **Information Sources:** methods to collect evidence about user preferences
  - **Family of Methods:** how information is combined/processed to generate recommendations
  - **Presentation & Interface**: how recommendations are displayed to users.

Online Identity Management

- Identity management is key for recommender systems.
- Online identity can be anonymous, pseudonymous, or attributed.
- Anonymous - no personal data is collected.
- Pseudonymous - users are identifiable by a chosen identifier.
- Attributed - users are identifiable by their real personal data.
- Choice of identity management impacts personalization.

Sources of information

- Explicit ratings on a numeric/ 5-star/3-star etc. scale
- Explicit binary ratings (thumbs up/thumbs down)
- Implicit information, e.g.,
    - who bookmarked/linked to the item?
    - how many times was it viewed?
    - how many units were sold?
    - how long did users read the page?
- Item descriptions/features
- User profiles/preferences

Method's type:

- Content-based filtering
  - recommendations based on item descriptions/features, and profile or past behavior *of the "target" user only*
- Collaborative filtering
  - recommendations based on past behavior of *other users as well as the target user*
- Hybrid methods

About The Music Genome Project®

The Music Genome Project powers Pandora. It's the most comprehensive analysis of music ever undertaken.

For over a decade, we've been gathering musical knowledge to bring you the best, most personalized listening experience out there.

We believe each individual has a unique relationship with music – no one has tastes that are exactly the same. So delivering a great experience to every listener requires a broad and deep understanding of music.

Our team of trained musicologists has been listening to music across all genres and decades, including emerging artists and new releases, studying and collecting musical details on every track– 450 musical attributes altogether.

The result of all our work is a personalized listening experience filled with both old favorite and new discoveries.

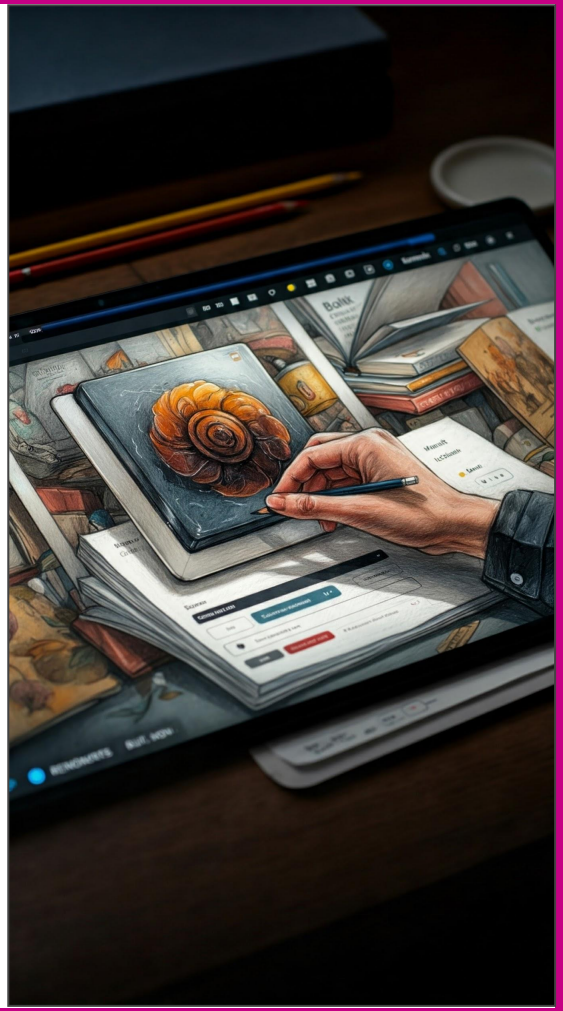
Content-filtering recommenders

- e.g., Pandora content-based recommendation based on the music genome project
- Overall operation: categorize items, or identify items with similar features; then recommend either
  - categories that match stated user profile
  - items similar to others the target user has liked/bought etc.

Content-based filtering

Example:

1. Use the common words as a similarity measure.
2. Recommend “closest” item to liked items.
3. Content filtering similarity measures are domain-specific.



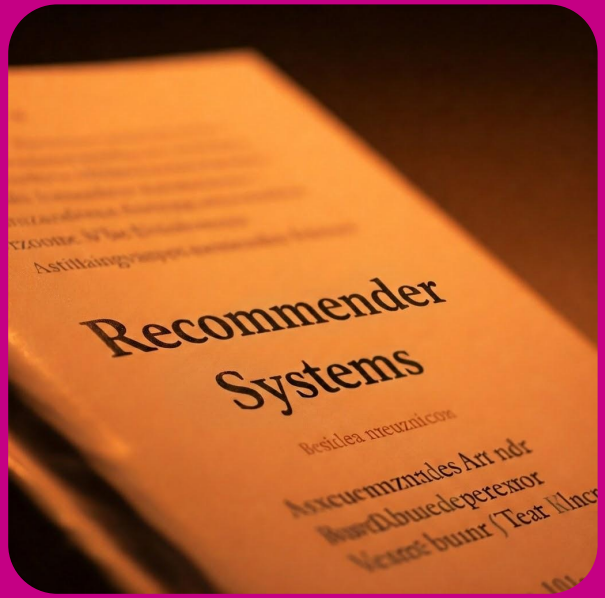
Collaborative filtering

- Main idea: users with similar tastes will tend to like similar items
- Use implicit/explicit ratings to:
  - find users similar to the current target user and recommend items they like
  - or, find item Y similar to item X for which most users who liked X like Y
  - or more complex approaches to learn a preference model from ratings
- Do not rely on domain-specific inputs-- basic algorithms can be applied to any CF setting

Hybrid methods

- Combine both content-based and collaborative filtering
- e.g., web search engines use keyword frequency metrics as well as link frequency to come up with a page list

Interface and Presentation



- Personalized/non-personalized recommendations
- Filter out bad items or show most recommended items first?
- Add explanations e.g. "recommended because you bought.."
- Other feedback to users about how much they have rated
- Quick ways to bootstrap new users

Business Models

- How is the recommendation site supported?
- Value-addition attached to a purchase/circulation service
- Advertisements
- Paid for by content owners
- Any others?

The Netflix Challenge

- In 2006 Netflix released 100 million anonymized ratings. Kept a set of about 1 Million ratings as a secret “test”
- Challenge: come up with algorithms to accurately predict “test” ratings
- Goal: 10% improvement over Netflix’s own algorithm
- On September 18, 2009, Netflix announced team "BellKor's Pragmatic Chaos" as the prize winner (a test RMSE of 0.8567),

RS trends

1. Interactive and Conversational Recommender Systems

- Moving from static recommendations to **dynamic, dialogue-based interactions**.
- Systems that:
  - Ask clarifying questions to refine preferences.
  - Provide **multi-turn conversations** for improved personalization.
- Examples: Chatbots, virtual assistants, voice-based recommendations.

2. Improving Offline Evaluation of RecSys

- Addressing the **gap between offline metrics and real-world performance**.
- Trends include:
  - Simulation of user behavior (e.g., synthetic user models).
  - Incorporating **user feedback loops**.
  - Better evaluation frameworks for diversity, serendipity, and novelty.

RS trends

3. Fairness, Bias, and Diversity in Recommendations

- **Key challenges:**
  - Avoiding biases that disproportionately favor certain users, items, or groups.
  - Breaking **filter bubbles** to expose users to diverse perspectives.
- Strategies:
  - Ensuring **algorithmic fairness** and transparency.
  - Balancing personalization with diversity.

4. Context and Temporal-Aware Recommendations

- Incorporating:
  - **Contextual factors**: location, device, weather, time of day.
  - **Temporal dynamics**: changing user preferences over time (short-term vs. long-term interests).
- Models leveraging:
  - **Context-aware approaches** (e.g., matrix factorization with context).
  - Sequential and time-aware deep learning methods.

RS trends

5. Privacy and Security in RecSys

- Growing focus on protecting user data in the age of personalized systems.
- Trends include:
  - Federated learning: Training models without centralizing user data.
  - Differential privacy: Ensuring individual data cannot be re-identified.
  - Secure multiparty computation for privacy-preserving recommendations.

6. Multi-Modal Recommendation

- Utilizing diverse **data modalities** (e.g., text, images, audio, video) for richer recommendation insights.
  - Example: Recommending movies using trailers, user reviews, and posters.
- Techniques:
  - Vision-language models (e.g., CLIP).
  - Joint embedding spaces for multi-modal learning.

RS trends

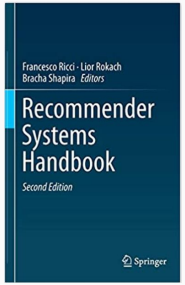
7. Generative AI in RecSys

- Leveraging generative models (e.g., **GPT**, **BERT**) to:
  - Improve **content-based recommendations** (e.g., product descriptions, personalized ads).
  - Generate synthetic user-item interactions for cold-start problems or evaluation.
- Examples: ChatGPT-based conversational recommenders.

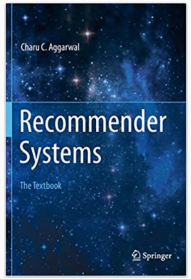
8. Sustainability in RecSys

- Designing energy-efficient recommendation models to reduce the **environmental impact** of large-scale systems.
  - Focus on lightweight architectures, pruning, and quantization.
- Encouraging recommendations that promote **sustainable consumption** (e.g., eco-friendly products).

Text books



Recommender Systems Handbook. Francesco Ricci,Lior Rokach, Bracha Shapira, Springer, 2015.



Recommender Systems: The Textbook. Charu C Aggarwal, Springer, 2016.

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Grado en Ciencia e Ingeniería de Datos

based on materials from Rahul Sami, and others.

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